

D3 7

16-20

$$M = h_c$$

$$V = v_i$$

	53	53,5	54	54,5	55	n_{h_c}
3 2 2	—	1	3	—	1	5
3 2 4	2	2	—	—	—	4
3 2 6	—	—	2	3	—	5
3 2 8	3	1	2	—	—	6
3 3 0	2	—	1	—	—	3
<u>n_{v_i}</u>	7	4	8	3	1	

$$N = \sum n_{v_i} = \sum n_{h_c} = 23$$

$$\bar{V} = \frac{1}{N} \sum V_i n_{v_i} = \frac{1}{23} \cdot 1235.5 \approx 53,72$$

$$\bar{H} = \frac{1}{N} \sum H_i n_{h_c} = \frac{1}{23} \cdot \del{3414} 7494 \approx 325,83$$

$$D(V) = \bar{V}^2 - (\bar{V})^2 = \frac{1}{23} \cdot 66375,75 - 2885,56 = 0,39$$

$$D(H) = \bar{H}^2 - (\bar{H})^2 = \frac{1}{23} \cdot 2447908 - 206762,67 = 7,47$$

$$\overline{VH} = \frac{1}{n} \sum V_i H_i = 902546/23 \approx 77502$$

$$K_{VH} = \overline{VH} - \bar{V} \cdot \bar{H} = 77502 - (53,72 \cdot 325,89) =$$

$$= 77502 - 17503,59 = -7,59$$

$$r_{VH} = \frac{K_{VH}}{\sqrt{D(V)D(H)}} \approx -0,63$$

i	X _i	Y _i	X _i ²	Y _i ²	X _i · Y _i
1	30	45	900	2025	1350
2	35	25	1225	625	875
3	37	48	1369	2304	1776
4	38	52	1444	2704	1976
5	41	54	1681	2916	2214
6	48	57	2304	3249	2736
7	50	59	2500	3481	2950
8	55	60	3025	3600	3300
Σ	328	394	14040	20256	76607

$$n = 8$$

$$\bar{x} = 328/8 = 41$$

$$\bar{y} = 394/8 = 49,25$$

$$D[X] = 24040/8 - 41^2 = 74$$

$$D[Y] = 20256/8 - 49,25^2 = 206,44$$

$$K_{xy} = 17660/8 - (41 \cdot 49,25) = 55,875$$

$$r_{xy} = \frac{K_{xy}}{\sqrt{D[X]D[Y]}} = \frac{55,875}{88,75} \approx 0,63$$

$$\alpha = 0,05$$

$$df = n - 2 = 6$$

$$t_{\text{emp}} = \frac{r_{xy} \sqrt{n-2}}{\sqrt{1-r_{xy}^2}} = \frac{0,63 \sqrt{6}}{\sqrt{1-0,63^2}} \approx 7,99$$

$$t_{\text{кр}} |0,05; 6| = 2,447$$

$t_{\text{emp}} < t_{\text{кр}}$: гипотезу про равенство
принято.

$$y = \beta_0 + \beta_1 x$$

$$\beta_1 = \frac{K_{xy}}{D(x)} = \frac{55.875}{79} \approx 0.755$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x} = 49.25 - 0.755 \cdot 67 = 78.295$$

$$y = 78.295 + 0.755x$$

$$R^2 = r_{xy}^2 = 0.63^2 \approx 0.397 = 39.7\%$$

